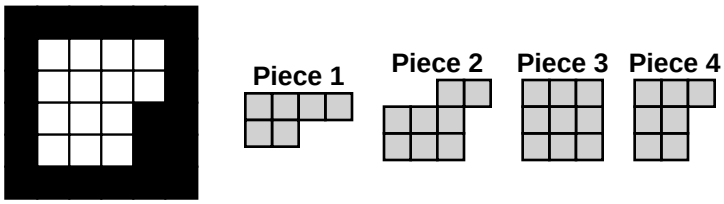
Math Kangaroo Practice Exam - Grades 3-4

20 Randomly Selected Practice Questions (Similar to Contest Problems — Never Originals)

Question 1. Anna, Ben, Clara, and David each own a different pet: a rabbit, a hamster, a turtle, and a canary. Ben and Clara both own furry pets, David owns a bird, and Ben does not have the hamster. Which statement is **not** true?

A) Ben has a rabbit **B)** David has a canary **C)** Clara has a rabbit
D) Anna has a turtle **E)** Clara has a hamster

Question 2. The picture shows a 6 × 6 square with some cells black and the rest white. We want to cover the white area completely without overlaps using two of the numbered pieces (1, 2, 3, 4). Which two pieces should we choose?

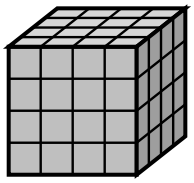


A) 1 and 2 **B)** 1 and 3 **C)** 2 and 4 **D)** 3 and 4 **E)** 2 and 3

Question 3. The odometer of a bus shows **478659**, where every digit is different. After how many more kilometres will all six digits be different again?

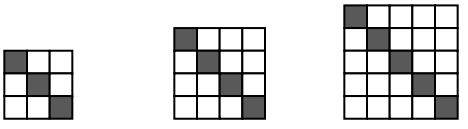
A) 1 **B)** 11 **C)** 21 **D)** 29 **E)** 31

Question 4. A 4 cm × 4 cm × 4 cm cube is painted grey on the outside and cut into 64 unit cubes of side 1 cm. How many of the unit cubes have **exactly 1** face painted?



A) 6 **B)** 16 **C)** 24 **D)** 36 **E)** 48

Question 5. The three diagrams show a pattern where the main diagonal is shaded grey (with 6, 12, and 20 white cells). How many white cells will the **next** diagram in the pattern have?



A) 24 **B)** 30 **C)** 32 **D)** 36 **E)** 42

Question 6. A wooden board is 20 dm, 80 cm, and 60 mm long. A carpenter saws off a piece that is 45 cm long. How long is the remaining board?

A) 2 m 86 cm **B)** 3 m 31 cm **C)** 2 m 95 cm
D) 2 m 35 cm **E)** 2 m 41 cm

Question 7. How many different letters appear in **both** the word **BALLOON** and the word **BANANA**?

A) 2 **B)** 3 **C)** 4 **D)** 5 **E)** 6

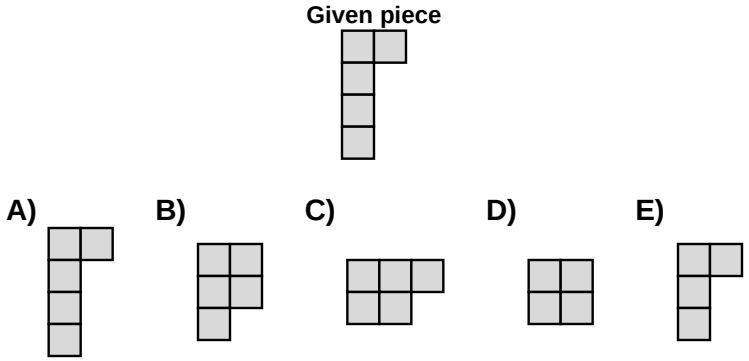
Question 8. Clara, who is **younger** than Leo by one year and one day, was born on January 1, 2015. What is the date of Leo's birthday?

A) January 2, 2016 **B)** January 2, 2014 **C)** December 31, 2013
D) December 31, 2014 **E)** December 31, 2016

Question 9. How many digits are needed to write down all whole numbers from **1 to 50**?

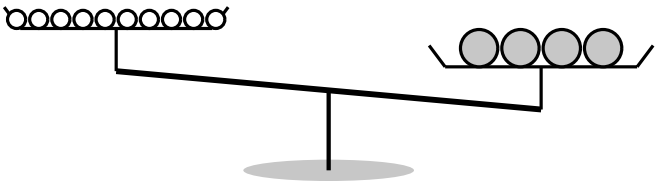
A) 50 **B)** 91 **C)** 99 **D)** 100 **E)** 101

Question 10. Which of the shapes (A through E) can be joined together with the given shape to form a solid rectangle without gaps or overlaps?



Question 11. On one plate there are 10 cherries (small circles) and on the other 4 apples (gray circles). When we put 2 more apples on the cherry plate, the balance is level. One apple weighs the same as:

A) 2 cherries **B)** 3 cherries **C)** 4 cherries
D) 5 cherries **E)** 10 cherries



Question 12. Leo built a tunnel 3 cubes deep with a 5 × 5 outer frame and a 3 × 3 tunnel opening (fig.1). He then used some of those cubes to build a 3-layer stepped pyramid with 5 × 5, 3 × 3, and 1 × 1 layers (fig.2). How many cubes from the tunnel were left over?

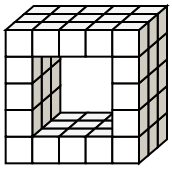


fig.1

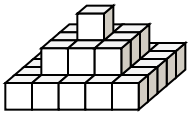
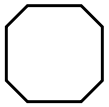
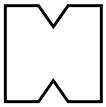
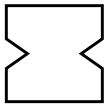


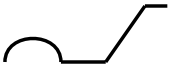
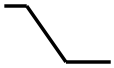
fig.2

A) 13 **B)** 15 **C)** 18 **D)** 21 **E)** 23

Question 13. A square piece of paper is folded in half vertically and then in half horizontally to form a smaller square. **Exactly one** of the four corners of this folded square is cut off with a straight snip, and the paper is unfolded. Which of the 5 designs below **cannot** be obtained?

A)  **B)**  **C)** **D)**  **E)**

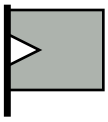
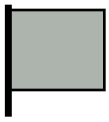
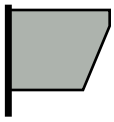
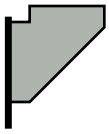
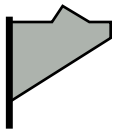
Question 14. Here is the skyline of a castle. Which piece **cannot** belong to it?

A) **B)**  **C)**  **D)** **E)**

Question 15. What is the smallest number greater than **3059** that has the same sum of digits as 3059?

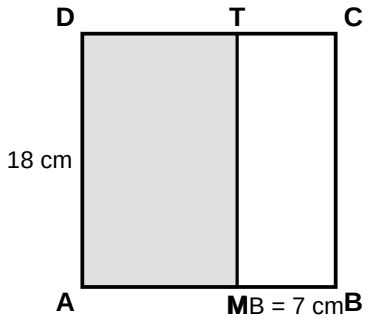
A) 3060 **B)** 3067 **C)** 3069 **D)** 3068 **E)** 3158

Question 16. Maya draws 5 silhouettes of a rectangular flag flying in the wind. One silhouette is impossible unless the flag is ripped along the pole. Which one is it?

A)  **B)**  **C)**  **D)**  **E)** 

Question 17. The side of square *ABCD* is 18 cm. A line segment *MT* divides the square into two rectangles. The length of segment *MB* is 7 cm. How much greater is the perimeter of square *ABCD* than the perimeter of rectangle *AMTD*?

A) 12 cm **B)** 14 cm **C)** 16 cm **D)** 18 cm **E)** 22 cm

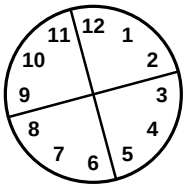


Question 18. It takes exactly **49 digits** to print all the page numbers of a storybook (starting from page 1). How many pages are in the storybook?

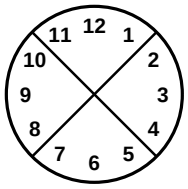
A) 20 **B)** 24 **C)** 25 **D)** 29 **E)** 40

Question 19. The face of a clock is cracked into 4 wedge-shaped pieces meeting at the center. When ordered from smallest to largest, the sums of the pieces increase by 3 each time (15, 18, 21, and 24). Which clock face shows how it was cracked?

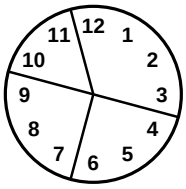
A)



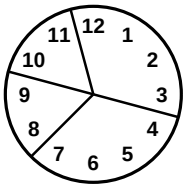
B)



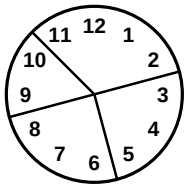
C)



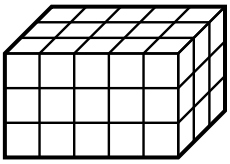
D)



E)



Question 20. Anna built this brick (5 cubes wide, 3 cubes tall, 3 cubes deep). The outside is completely red and the cubes inside are blue. How many **red** cubes did she use?



A) 30 **B)** 36 **C)** 39 **D)** 42 **E)** 45

Answer Key - 20-Question Practice Exam

For generated/2026-10-19-19-27-51.pdf (Randomly sampled from question-pool.txt)

Solutions & kid-friendly explanations

Question 1 (inspired by 2002 #18): C David has the bird (canary). The two furry pets are the rabbit and the hamster, belonging to Ben and Clara. Since Ben does not have the hamster, Ben has the rabbit and Clara has the hamster. That leaves the turtle for Anna. Thus, 'Clara has a rabbit' (C) is not true.

Question 2 (inspired by 2003 #18): A The white area has 14 squares. Placing Piece 1 (6 squares) at the top of the white region leaves an 8-square stepped shape at the bottom that matches Piece 2 exactly.

Question 3 (inspired by 2002 #24): E From 478659, numbers 478660–478669 repeat 6, 478670–478679 repeat 7, and 478680–478689 repeat 8. The next number is 478690, whose digits (4, 7, 8, 6, 9, 0) are all different! That happens after 478690 - 478659 = 31 km.

Question 4 (inspired by 2007 #17): C On each of the 6 faces of the 4 × 4 × 4 cube, the inner 2 × 2 = 4 squares (not touching any edge) have 1 painted face. Across all 6 faces, that is 6 × 4 = 24 unit cubes.

Question 5 (inspired by 2007 #21): B The grids are 3×3 (9 - 3 = 6 white), 4×4 (16 - 4 = 12 white), and 5×5 (25 - 5 = 20 white). The next grid is 6×6 = 36 cells with 6 diagonal grey cells, giving 36 - 6 = 30 white cells.

Question 6 (inspired by 2003 #10): E Convert the initial length to centimetres: 20 dm = 200 cm, 80 cm = 80 cm, and 60 mm = 6 cm, totaling 200 + 80 + 6 = 286 cm. Subtracting 45 cm leaves 286 - 45 = 241 cm = 2 m 41 cm.

Question 7 (inspired by 2007 #3): B The distinct letters in BANANA are B, A, and N, and all three of them also appear in BALLOON. Counting distinct shared letters gives 3 common letters (B, A, and N).

Question 8 (inspired by 2007 #12): C Because Clara is younger than Leo, Leo is older and was born 1 year and 1 day BEFORE Clara. One year before January 1, 2015 is January 1, 2014, and one day before that is December 31, 2013.

Question 9 (inspired by 2007 #23): B The 9 one-digit numbers (1 to 9) use 9 × 1 = 9 digits. The 41 two-digit numbers (10 to 50) use 41 × 2 = 82 digits. Altogether, 9 + 82 = 91 digits are needed.

Question 10 (inspired by 2007 #8): A The given shape has column heights 4 and 1 (5 squares). Shape A is identical; turned upside-down it has column heights 1 and 4. Joining them gives columns of height 4+1=5 each, creating a 5 × 2 rectangle.

Question 11 (inspired by 2002 #9): D 10 cherries + 2 apples = 4 apples, so 10 cherries = 2 apples, so 1 apple = 5 cherries.

Question 12 (inspired by 2002 #20): A Each of the 3 layers of the tunnel has 5×5 - 3×3 = 25 - 9 = 16 cubes, so the tunnel has 3 × 16 = 48 cubes. The pyramid uses 25 + 9 + 1 = 35 cubes. That leaves 48 - 35 = 13 cubes unused.

Question 13 (inspired by 2007 #24): C Cutting a single corner of the twice-folded square can cut the 4 outer corners (A), the top and bottom edge midpoints (B), the left and right edge midpoints (D), or the center (E). To get notches on all four edges at once (C), you would have to cut two different corners of the folded square, so C cannot be obtained with one corner cut.

Question 14 (inspired by 2002 #7): E The castle wall has TWO notches at the top. Piece E shows a whole tower top with only one notch.

Question 15 (inspired by 2007 #4): D The digit sum of 3059 is 3 + 0 + 5 + 9 = 17. Increasing the tens digit by 1 (from 5 to 6) and decreasing the ones digit by 1 (from 9 to 8) gives 3068, whose digits also sum to 3 + 0 + 6 + 8 = 17.

Question 16 (inspired by 2002 #15): A In picture A, there is a triangular gap right along the middle of the flagpole where the flag is supposed to be attached. Since the left edge is fixed to the pole, that gap cannot appear unless the flag is torn.

Question 17 (inspired by 2002 #6): B Since AB = 18 cm and MB = 7 cm, AM = 18 - 7 = 11 cm. Perimeter of square = 4 × 18 = 72 cm. Perimeter of rectangle AMTD = 2 × (18 + 11) = 58 cm. Difference = 72 - 58 = 14 cm (which is simply 2 × MB = 2 × 7 = 14 cm).

Question 18 (inspired by 2003 #16): D Pages 1 to 9 take 9 digits, leaving 49 - 9 = 40 digits. Each two-digit page takes 2 digits, which gives 40 ÷ 2 = 20 two-digit pages. Adding the 9 single-digit pages gives 9 + 20 = 29 pages.

Question 19 (inspired by 2002 #11): C In clock C, the four wedges are {4, 5, 6} = 15, {12, 1, 2, 3} = 18, {10, 11} = 21, and {7, 8, 9} = 24. Sorted in order, the sums are 15, 18, 21, and 24.

Question 20 (inspired by 2003 #9): D Total = 5 x 3 x 3 = 45. Inside (blue) = 3 x 1 x 1 = 3. Red = 45 - 3 = 42.